

LUCAS PALOMINO

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EDUCATION	University of Wisconsin-Superior	Superior, WI
	B.S. in Computer Science (Cum Laude)	Sep 2021 - May 2025
	B.S. in Environmental Studies (Summa Cum Laude)	
	Certificate: Geographic Information Systems Certificate: Sustainable Enterprise Management	
EXPERIENCE	Research Intern	July 2025 - Present
	University of Wisconsin-Madison	Madison, WI
	• Conducting data analysis on YOLO object detection outputs to assess cattle distribution and compliance patterns in embargoed areas of the Amazon region • Leading team of 3 undergraduates in GIS-based spatial analysis to map and assess PFAS contamination patterns across U.S. farmlands	
	IT Support Specialist	May 2022 - May 2025
	University of Wisconsin-Superior	Superior, WI
	• Provided technical support and troubleshooting for 1,400+ IT issues in-person and remotely • Refined verbal and written communication skills by answering calls and documenting solutions via ticket system • Manage an equipment checkout database consisting of 323 devices, ensuring timely availability and returns	
	Renewable Energy Intern	Jun 2024 - Aug 2024
	LevelTen Energy	Seattle, WA
	• Received training and mentorship on project development, grid infrastructure, and electricity markets from energy professionals at sponsoring companies including LevelTen, Energy GPS, Clearway, GridStor, SB Energy, and Google • Designed 5 SQL models for LevelTen's BigQuery data warehouse, aggregating over 42,000 rows of data • Developed and tested database queries following SOPs and version controlling with Bitbucket	
	Programming Student Grader	Jan 2022 - Dec 2022
	University of Wisconsin-Superior	Superior, WI
	• Reviewed 311 coding assignments for an introductory-level programming class, practicing strong attention to detail • Managed grading deadlines and priorities to provide timely feedback to students	

PROJECTS	Correlating Corn-Soybean Rotations and Soil Nutrients Link		Jan 2025 - May 2025
	- Expanded previous GIS workflow to examine corn–soybean rotation effects on soil carbon, nitrogen, and phosphorus across the U.S. Corn Belt - Integrated NRCS soil pedon data (C, N, P) with USDA’s Cropland Data Layer (CDL) and analyzed statistical trends in Python - Found that frequent corn–soybean rotations were associated with modest increases in soil carbon and nitrogen, but showed no clear relationship with phosphorus		
	Automated GIS Workflow for Agricultural Analysis Link		Aug 2024 - Dec 2024
OUTREACH	- Developed an open-source workflow that automatically identifies soybean-corn rotations across the US - Built a REST API script to dynamically retrieve GeoJSON, raster, and vector data from the USDA’s service directory - Automated 15 spatial operations using GRASS Python modules, creating maps and statistics on the fly		
	Predicting Land Suitability for Trees in Brazil with ML Link		Aug 2024 - Dec 2024
	- Assembled geospatial dataset of 2,585 tree locations with biophysical and climatic variables from multiple data sources - Trained decision tree and neural network classifiers in Scikit-learn to predict which of 7 native tree species would thrive at given locations - Evaluated models using accuracy, precision, recall, F1 scores, and AUC-ROC, identifying class imbalance as a key area for improvement		
VOLUNTEER	Community Lead College to Climate		May 2024 - May 2025
	- Cultivated relationships with 100+ climate employers, university career centers, and student organizations - Organized 6 events, including 4 monthly speaker events, a climate finances panel, and a NYC Climate Week event - Presented community data analytics to career center staff from 19 universities including MIT, Columbia, NYU, Cornell, UC Berkeley, Duke, UPenn, and Brown		
	Sustainability Club President University of Wisconsin-Superior		Jan 2022 - Jun 2023
PRESENTATIONS	- Managed \$15,000 in Sustainability Club assets and delegated a team of 10 officers - Launched the OZZI reusable container program, providing an alternative to single-use containers at UW-Superior dining options - Attended the MIT Sustainability Summit 2024 workshop on food system challenges		
	Sustainability Liaison University of Wisconsin-Superior		Sep 2021 - May 2022
	- Informed the student senate about campus sustainability initiatives - Explored the idea of a “Green Fee” to fund sustainability efforts at UW-Superior		
VOLUNTEER	Cloud Native Geospatial Conference		Apr 2025
	Registration & Logistics		Salt Lake City, UT
VOLUNTEER	NY Climate Week: Earth Observation for Climate		Sep 2024
	Logistics		New York City, NY
PRESENTATIONS	1. Northeastern GIS User Group Spring 2025 Meeting		May 2025
	<i>Designing an Open-Source GIS Workflow for Automating the Identification Of Soybean-Corn Rotations Across the US</i>		
PRESENTATIONS	2. Universities of Wisconsin Research in the Rotunda		Apr 2025
	<i>Automated GIS Workflow for Agricultural Analysis (Poster Session)</i>		

AWARDS AND HONORS	Math and Computer Science Scholarship , University of Wisconsin-Superior	S2024
	Tommy Thompson Leadership Scholarship , University of Wisconsin-System	S2023
	Academic Scholarship , University of Wisconsin-Superior	S2022

SKILLS	Languages: English, Spanish, Portuguese, French (elementary proficiency)
	Programming Languages: Python, C#, C++, Java, JavaScript
	Structured Languages: SQL, HTML, CSS, XML, LaTeX
	Software: ArcGIS Pro, QGIS, Git, AutoCAD, BigQuery, Excel